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Digital entertainment is the subject of focus for most computer component manufacturers and the humble PC is slowly, but steadily, morphing into a true multimedia capable machine. With the release of new motherboard chipsets, capable of decoding Blu-Ray content on the fly, PCs' multimedia capabilities are bound to get a shot in the arm. This new breed of PCs will be capable of running high resolution movies, playing graphically intensive games, ripping hi-definition audio without breaking a sweat.

These new capabilities demand higher capacity and better performing storage sub-systems. Getting 500 GB or 1 TB of hard drives will no longer need an excuse but be a necessity. The good news being, storage space is getting cheaper. A year ago, a 1 TB drive would have easily set you back by around Rs 15,000 but today you can buy the same or newer generation hard drives for approximately Rs 9,000. If that's not convincing enough, a 640 GB drive retails for as little as Rs 4,000 and makes a terrific buy.

Put two of them together and you have enough space to store as much hi-def content as you wish.

But digital data hardly sits idly on any hard drive; it's always shared and moving around from one PC to another. Internal drives may be cheap, but when it comes to using them for exchanging data they are cumbersome to use and not the ideal solution. By now, most of us have pen drives (Flash drives) which are often used to transfer data from one PC to another. But will these flash drives live up to the challenge when faced with hi-def content? Maybe not; they are slow, and expensive. A good external drive is the right solution as it offers the advantages of both internal hard drives and flash drives in one single package.

External hard drives are portable, reliable, are fast enough to transfer data and work out cheaper when you take into consideration the price per GB. Today, an 8 GB flash drive retails for around Rs 1,500 — in other words it works out to Rs 190 per



LaCie Little Disk 320 GB

GB. On the other hand, a 120 GB external hard drive is available for Rs 3,500, if you do the math, the cost of 1 GB works out to Rs 32 only.

While many might argue about the difference between the form factors of flash drives and external hard drives, one look at the new generation of external hard drives will be enough to put your contention to rest.

External drives have evolved from the dirty DIY packages which were rampant in the early years to sleek, stylish and affordable devices. External hard drives are classified into two categories: portable and desktop. As the name suggests, portables

How We Tested

Test Rig Configuration

Our test machine consisted of an Intel Core 2 Extreme X6800 processor running at 2133 MHz on a Gigabyte GA-965G-DS3 motherboard-based on the Intel 965G chipset, with 1 GB Kingston HyperX DDR2 533 MHz RAM and a Seagate Barracuda 750 GB SATA II drive. A fresh installation of Windows XP with SP2 was done and the latest versions of the hardware drivers were installed along with DirectX 9.0c. We used Intel's latest X25 SSD drive as the source and destination drive to test the real world Read / Write performance. All drives were hooked on an USB 2.0 port, except for FireWire drives which were hooked to a FireWire 400 port.

Test Procedure

The test suite consisted of synthetic benchmarks such as HD Tach 3.0.1.0 and Si Soft Sandra. Real world test consisted of file transfer test, between two drives and within two partition of the same drive. We also noted down important features such as dimension, weight (for portable drives), types of interface offered, bundled accessories, software bundled etc. These features were then given appropriate weightage to get the final features score. In similar fashion we also gave weightages to

performance figures and calculated the performance score. Price per GB is an important criterion for hard drives and it was also taken into consideration along with features and performance score to declare the final winner.

Benchmarks

HD Tach 3.0.1.0

The fresh hard drive to be tested was connected to the test machine and not formatted and allowed to remain raw (unpartitioned). HD Tach RW 3.0.1.0 was then run and used to evaluate the hard drive.

SiSoft Sandra

The File system benchmark gives valuable insight into the low level performance of the hard drive. These score generally tally well with the actual figures and hence are important.

Real World File Copy Test

We took 1 GB of assorted and sequential files to mimic real world conditions and transferred them across two drives and with two partitions of the same test drive. For each run we noted down the time taken to complete the file copy command.

are the ones you carry around and share data, whereas desktop drives are generally used for taking backups. In this test we have drives in both categories from prominent players in the market, some old and some new entrants. Western Digital, LaCie, Freecom and Verbatim are here. Seagate was in the process of getting their new line up of external drives and hence couldn't participate in this test, but we will definitely review their products once they're released.

Portable External Drives

Features Design

Looks play and important role in attracting consumers, especially for mundane products such

the Passport drives. Verbatim's SmartDisk feels plasticity and the silver finish doesn't do anything to enhance the looks. LaCie's rectangular and boxy Little Disk, designed by Sam Hecht, is finished in translucent material, which looks good, but is a smudge magnet. In our opinion, the WD Passport drives stand out from the crowd.

Portability

Here, portability refers to compact dimensions. The form factor determines whether the device will be carried along or will spend most of the time on the desktop. Western Digital's Passport drives come top here with their sleek dimensions. LaCie's Little Disk comes second with dimensions on a par with WD; it could have done



Western Digital Passport Studio 320 GB

as external hard drives. Every manufacturer has done their bit to make them appealing, and have roped in known designers to create their products. LaCie, for example, has drives designed by the legendary Porsche Design studios. Western Digital (WD) was the first to style their drives, and their Passport series of drives is sleek, stylish and attractive. The new Passport Essential series offers drives in eleven colours, whereas their Elite and Studio have an anodized finish. Freecom's Tough Drive Pro was completely encased in a white protective covering; this looks good, but can't match

better with rounded edges though. Freecom's Tough Drive Pro, though sleek, is elongated, and will have trouble fitting in a pocket. Verbatim drives have a curved dome and the drives taper from front to back. Despite their relatively bulky shape, Verbatim drives are the lightest. In fact, we felt they had just sent us the shell with no drive inside. Freecom's Tough Drive Pro with its shock resistant armour tipped the scales at 200 grams — relatively heavy, but well protected. LaCie's Little Disk and WD's Passport were nearly identical in weight — they feel right, not too heavy.

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